NAMI / AC CURRENT MONITOR

Thread + BLE | SCT006 | 16 MiB local history

ENGINEERING PROTOTYPE — ISOLATED LOW-VOLTAGE BENCH USE

POWER

File: power.kicad_sch

ANALOG

File: analog.kicad_sch

DIGITAL

File: digital.kicad_sch

HARNESS

File: harness.kicad_sch

External sensor harness: SCT006 20A:25mA + permanent 5.00-ohm burden at the clamp.

No mains conductors terminate on this board. Battery/bench source connects at J2.

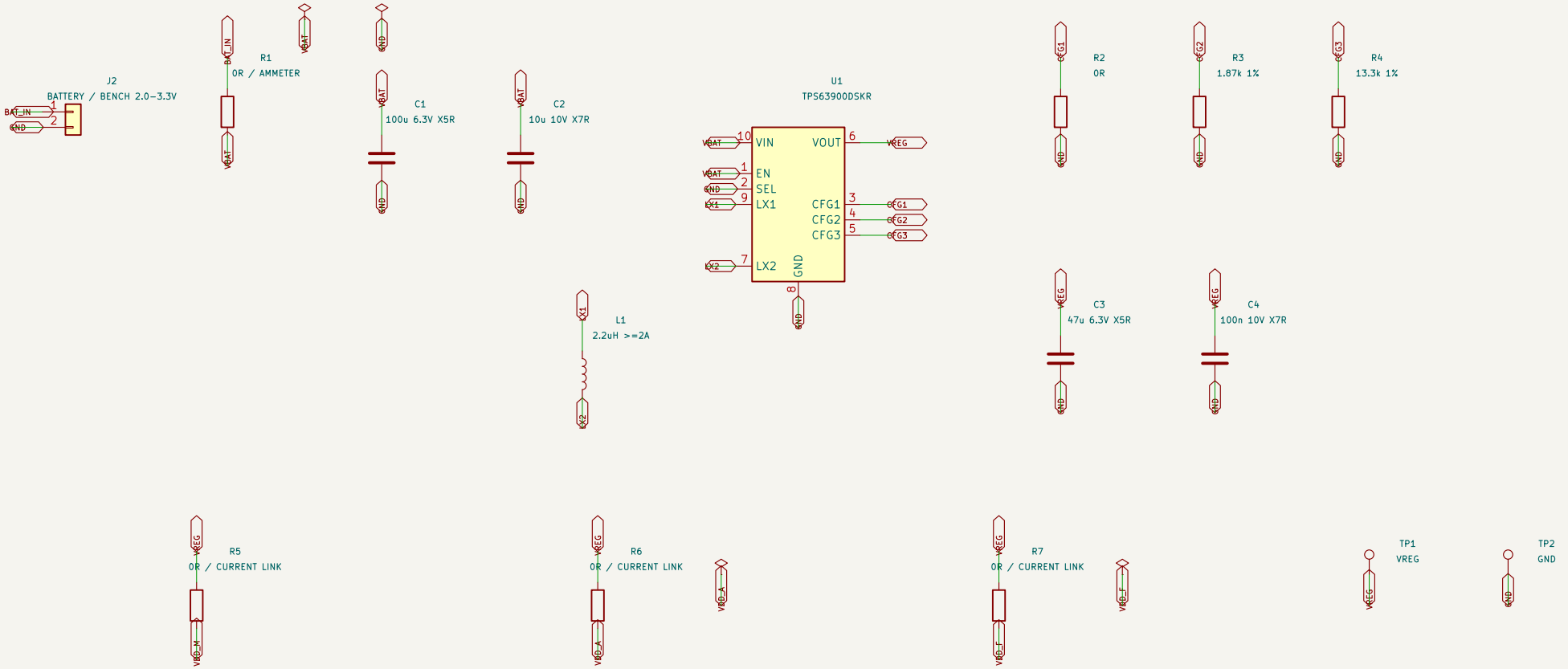
Schematic and routed PCB are review artifacts. See README for verification and release status.

Sheet: /
File: nami.kicad_sch

Title: Nami — Rev A bench prototype

Size: A3	Date: 2026-09-26	**Rev: A0**
KiCad E.D.A. 10.0.6		Id: 1/5

POWER / 3.0 V regulated / battery or bench supply

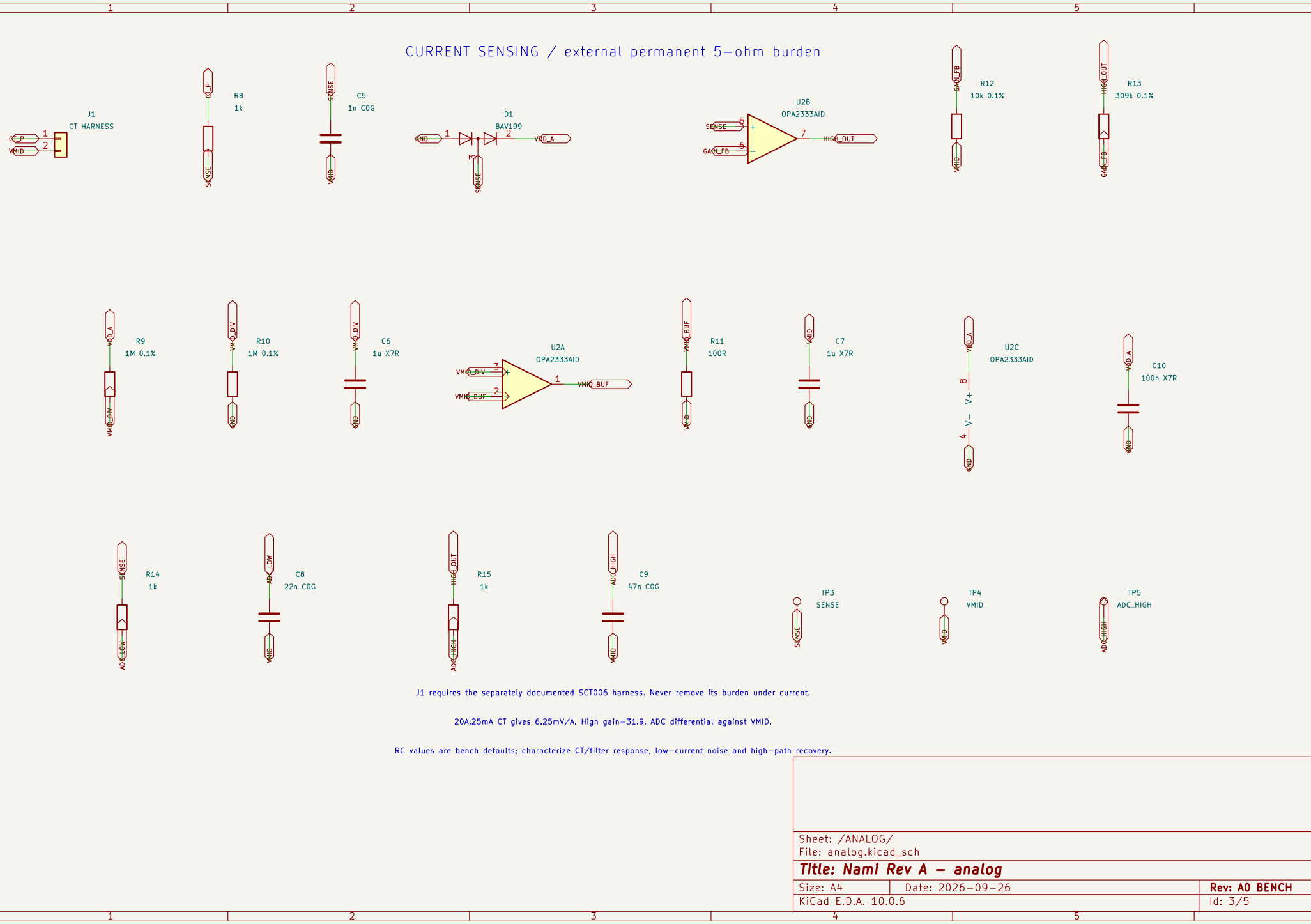


U1: SEL grounded; CFG3=13.3k selects 3.0V. CFG2=1.87k limits input to 25mA.

Single external source only. R1/R5/R6/R7 permit subsystem current measurements.

Battery chemistry: primary 3V cell, no charging. Validate pulse sag, startup and brownout.

Sheet: /POWER/		
File: power.kicad_sch		
Title: Nami Rev A – power		
Size: A4	Date: 2026-09-26	**Rev: A0 BENCH**
KiCad E.D.A. 10.0.6		Id: 2/5


CURRENT SENSING / external permanent 5-ohm burden
J1
CT HARNESS
CT_P
VMID
R8
1k
SENSE
C5
1n C0G
D1
BAV199
GND
VDD_A
U2B
OPA2333AID
SENSE
GAIN_FB
HIGH_OUT
R12
10k 0.1%
R13
309k 0.1%
R9
1M 0.1%
VMID_DIV
R10
1M 0.1%
C6
1u X7R
U2A
OPA2333AID
VMID_BUF
R11
100R
C7
1u X7R
U2C
OPA2333AID
V+
V-
C10
100n X7R
R14
1k
ADC_LOW
C8
22n C0G
R15
1k
ADC_HIGH
C9
47n C0G
TP3
SENSE
TP4
VMID
TP5
ADC_HIGH
J1 requires the separately documented SCT006 harness. Never remove its burden under current.
20A:25mA CT gives 6.25mV/A. High gain=31.9. ADC differential against VMID.
RC values are bench defaults; characterize CT/filter response, low-current noise and high-path recovery.
Sheet: /ANALOG/
File: analog.kicad_sch
Title: Nami Rev A – analog
Size: A4
Date: 2026-09-26
Rev: A0 BENCH
KiCad E.D.A. 10.0.6
Id: 3/5

THREAD / BLE / STORAGE / SERVICE

U3: normal-voltage mode; VDD, VDDH and DCCH tied together. VBUS grounded; D+/D- unused.

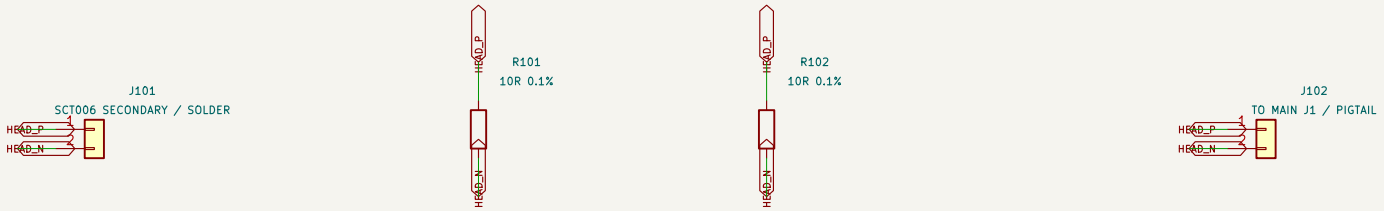
P0.02/03/04 = low/high/midpoint. Differential SAADC at +/-0.6V; start at 32768 samples/s per range.

Configure P0.18 as RESET in UICR. Enable REG1 DC/DC. Keep radio antenna copper-free on all layers.

Sheet: /DIGITAL/	
File: digital.kicad_sch	
Title: Nami Rev A — digital	
Size: A3	Date: 2026-09-26
KiCad E.D.A. 10.0.6	Id: 4/5

Rev: A0 BENCH

PERMANENT CT HARNESS / OFF-BOARD ASSEMBLY



SCT006 20A:25mA. R101 and R102 form a permanent 5.00-ohm burden.

J101 represents solder joints, NOT a removable connector. Insulate and strain-relieve the assembly.

J102 pin 1 -> board J1 pin 1; J102 pin 2 -> board J1 pin 2 (VMID).

Do not add a second burden on the main PCB. Verify factory secondary protection on the ordered CT.

This assembly never connects electrically to the monitored primary conductor.

Sheet: /HARNESS/		
File: harness.kicad_sch		
Title: Nami Rev A — harness		
Size: A4	Date: 2026-09-26	**Rev: A0 BENCH**
KiCad E.D.A. 10.0.6		Id: 5/5